Oxford College of Emory University

Chapter 15: Monetary Policy

Part 1: Money Demand, Interest Rates, and Aggregate Demand

ECON_OX 112 · Principles of Macroeconomics

Guilherme Paiva Pinto

Case A: March 15, 2020

- The pandemic threatened economic activity and jobs.

FOMC statement, March 15, 2020

Case A: March 15, 2020

- The pandemic threatened economic activity and jobs.
- The Fed cut its target range to 0–0.25% and announced at least $500 billion in Treasury purchases and $200 billion in mortgage-backed securities.

FOMC statement, March 15, 2020

Case A: March 15, 2020

- The pandemic threatened economic activity and jobs.
- The Fed cut its target range to 0–0.25% and announced at least $500 billion in Treasury purchases and $200 billion in mortgage-backed securities.
- Lower rates and bond purchases were meant to support lending, spending and employment.

FOMC statement, March 15, 2020

Case B: November 3, 2021

- Jobs were recovering, but inflation was elevated. The Fed then expected much of the inflation to be temporary.

FOMC statement, November 3, 2021

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- Jobs were recovering, but inflation was elevated. The Fed then expected much of the inflation to be temporary.
- It kept the target at 0–0.25%. Monthly purchases slowed to $70 billion in Treasuries and $35 billion in mortgage-backed securities.

FOMC statement, November 3, 2021

Case B: November 3, 2021

- Jobs were recovering, but inflation was elevated. The Fed then expected much of the inflation to be temporary.
- It kept the target at 0–0.25%. Monthly purchases slowed to $70 billion in Treasuries and $35 billion in mortgage-backed securities.
- It was reducing support gradually: bond holdings were still growing, just more slowly.

FOMC statement, November 3, 2021

Case C: June 15, 2022

- High inflation was the main concern.

FOMC, June 15, 2022

Case C: June 15, 2022

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- The Fed raised the target range to 1.50–1.75%, raised interest on reserve balances to 1.65%, and continued reducing bond holdings.

FOMC, June 15, 2022

Case C: June 15, 2022

- High inflation was the main concern.
- The Fed raised the target range to 1.50–1.75%, raised interest on reserve balances to 1.65%, and continued reducing bond holdings.
- Higher rates were meant to slow borrowing and spending, reducing inflation pressure.

FOMC, June 15, 2022

Case D: November 1, 2023

- Inflation remained too high.

FOMC statement, November 1, 2023

Case D: November 1, 2023

- Inflation remained too high.
- The Fed kept the target range at 5.25–5.50% and continued reducing its bond holdings.

FOMC statement, November 1, 2023

Case D: November 1, 2023

- Inflation remained too high.
- The Fed kept the target range at 5.25–5.50% and continued reducing its bond holdings.
- Keeping rates high maintained pressure on borrowing and spending. An unchanged rate can still restrain demand.

FOMC statement, November 1, 2023

Case E: September 18, 2024

- Inflation had eased, while job growth had slowed.

FOMC, September 18, 2024

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FOMC, September 18, 2024

Case E: September 18, 2024

- Inflation had eased, while job growth had slowed.
- The Fed cut the target by 0.50 percentage point to 4.75–5.00%. It continued reducing bond holdings.
- The cut was meant to reduce the risk of further job weakness while keeping inflation on a path toward 2%.

FOMC, September 18, 2024

Learning Objectives

- A · Understand and explain demand for money.
- B · Explain money-market equilibrium and the connection between money and interest rates.
- C · Interpret aggregate-demand shifts and demonstrate the effects of monetary policy.

15.1 Demand for Money

- Understand and explain demand for money.

Money or Bonds

- A household has $1,000 and needs $400 for bills next month.
- Money earns 0%. The bond pays interest but cannot be cashed in early in this exercise.
- Answer P1–P4: how much should the household keep available for payments?

The Opportunity Cost of Holding Money

Keep $600 for one year	Ending amount	Interest earned
Money at 0%	$600	$0
Bond at 1%	$606	$6
Bond at 6%	$636	$36

Rate of return = (ending amount − starting amount) ÷ starting amount

- For the 6% bond: $36 ÷ $600 = 6% per year.

SEC Investor.gov: bonds · SEC Investor.gov: certificates of deposit

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The Opportunity Cost of Holding Money

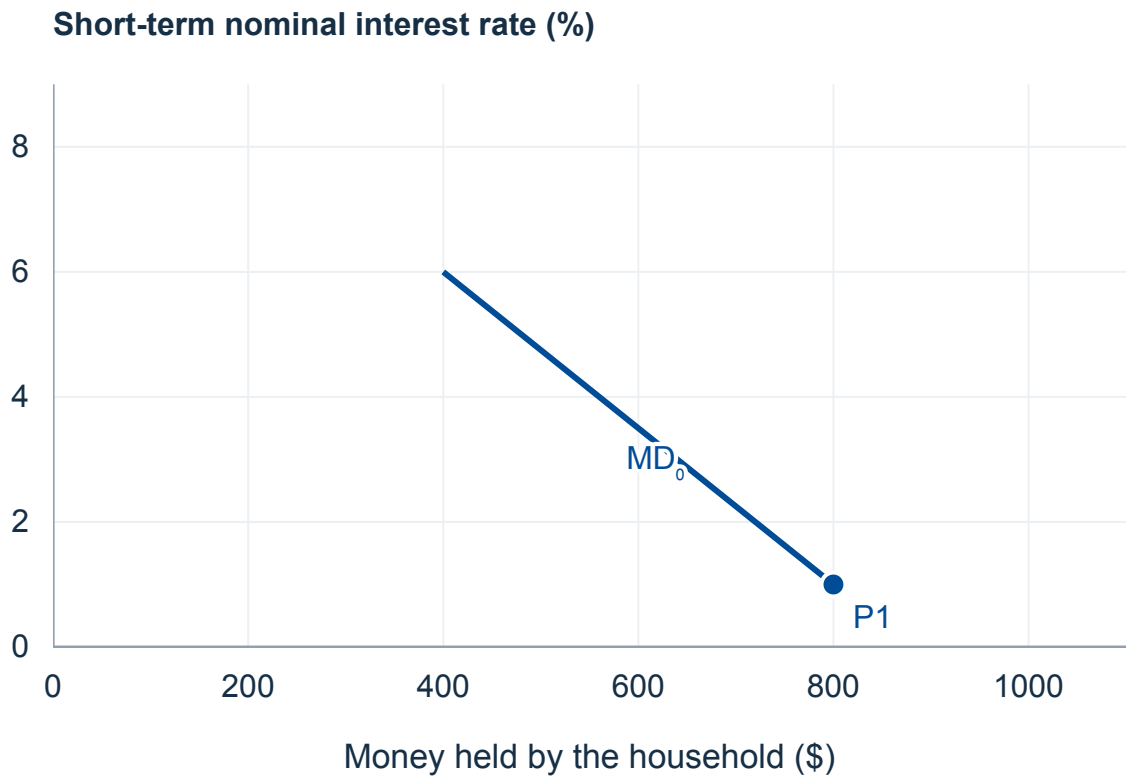
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- Keeping $600 in money means giving up that interest.
- When the bond rate rises from 1% to 6%, interest forgone rises from $6 to $36.

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Money Demand: P1–P3

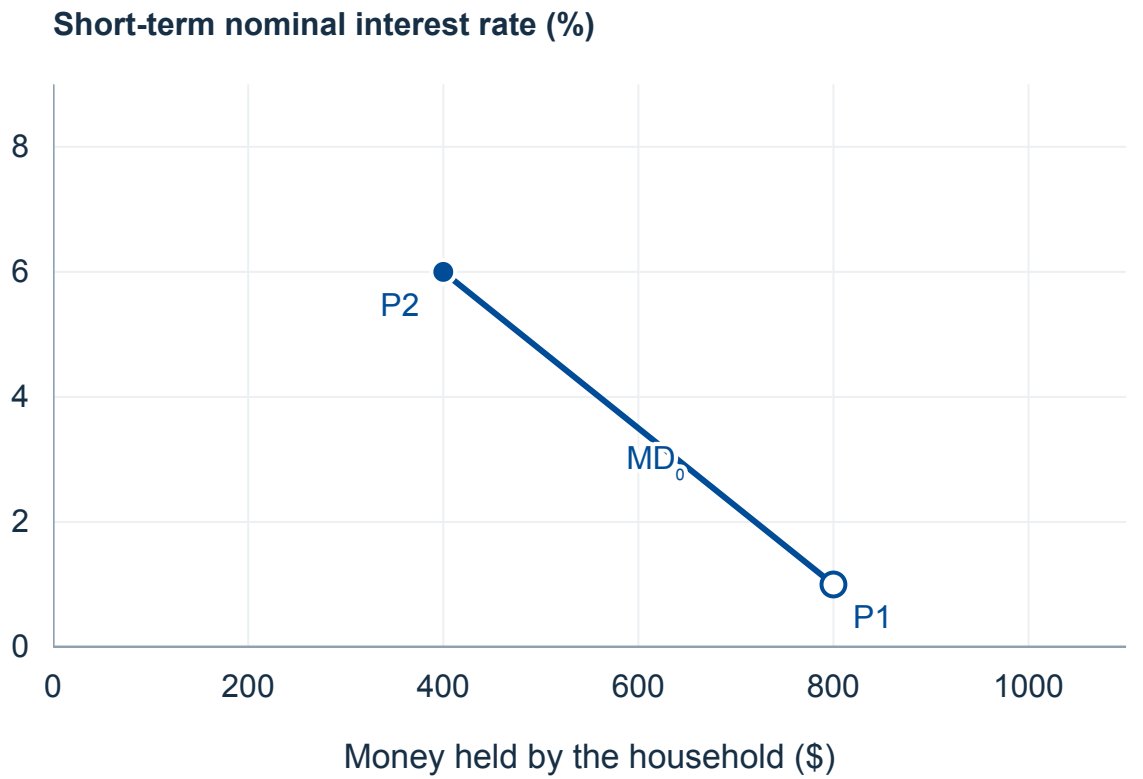


Illustrative choices from P1–P3; not class vote totals.

P3 changes the bills, not the interest rate.

P1: at 1%, the example household keeps $800 for spending.

Money Demand: P1–P3



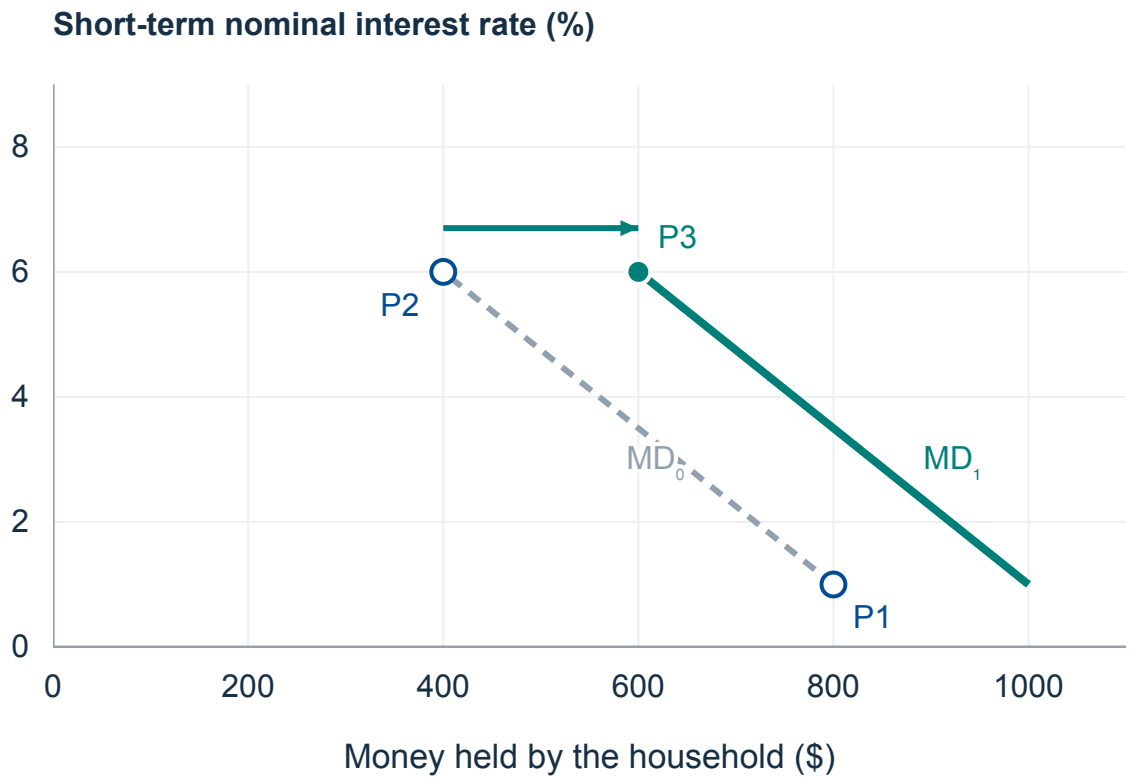
Illustrative choices from P1–P3; not class vote totals.

P3 changes the bills, not the interest rate.

P1: at 1%, the example household keeps $800 for spending.

P2: at 6%, it keeps $400. This is a movement along MD_0.

Money Demand: P1–P3



Illustrative choices from P1–P3; not class vote totals.

P3 changes the bills, not the interest rate.

P1: at 1%, the example household keeps $800 for spending.

P2: at 6%, it keeps $400. This is a movement along MD_0.

P3: bills rise from $400 to $600, but the rate stays at 6%. MD shifts right to MD_1.

Shifts in Money Demand

Price level

Real GDP

Credit and banking technology

Banking institutions

Shifts in Money Demand

Price level

Higher prices require more dollars for the same purchases. MD shifts right.

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Shifts in Money Demand

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More output means more purchases and payments. MD shifts right.

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Shifts in Money Demand

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More output means more purchases and payments. MD shifts right.

Credit and banking technology

Easier credit or faster payments can reduce cash buffers. MD may shift left.

Banking institutions

Shifts in Money Demand

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Credit and banking technology

Easier credit or faster payments can reduce cash buffers. MD may shift left.

Banking institutions

Rules that require larger account balances raise money demand. MD shifts right.

Money Demand Checkpoint

1. Interest rates rise; prices and spending stay the same. Movement along MD or a shift?

2. At the same interest rate, the same bills cost more dollars. Movement or shift?

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> Move up along MD: the quantity of money demanded falls.

2. At the same interest rate, the same bills cost more dollars. Movement or shift?

> MD shifts right: more money is needed for payments.

15.2 Money and Interest Rates

- Explain money-market equilibrium and the connection between money and interest rates.

The Federal Funds Rate

River Bank today	Amount
Reserves available	$2,000
Payments due tonight	$2,500
Overnight loan needed	$500

- River Bank can borrow the missing $500 overnight from another bank. The rate it pays is the federal funds rate (FFR).

Federal Reserve Board: policy goals and transmission · Federal Reserve Board: implementation and IORB

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- The FOMC sets a target range for the FFR.

Federal Reserve Board: policy goals and transmission · Federal Reserve Board: implementation and IORB

A Fed Bond Purchase

Account	The $1,000 purchase
Federal Reserve	Treasury bond +$1,000
River Bank	Treasury bond −$1,000 Reserves: $2,000

The Fed buys a $1,000 Treasury bond from River Bank.

The bond purchase adds reserves. New bank lending creates deposits.

Federal Reserve Board: implementation and IORB

A Fed Bond Purchase

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Federal Reserve	Treasury bond +$1,000 Bank reserves +$1,000
River Bank	Treasury bond −$1,000 Reserves: $2,000 + $1,000 → $3,000

The Fed buys a $1,000 Treasury bond from River Bank.

The Fed pays by adding $1,000 to the bank's reserves: $2,000 becomes $3,000.

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Overnight loan needed: $500 → $0

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The Fed pays by adding $1,000 to the bank's reserves: $2,000 becomes $3,000.

The bank can cover its $2,500 payment. It no longer needs the $500 overnight loan.

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Federal Reserve Board: implementation and IORB

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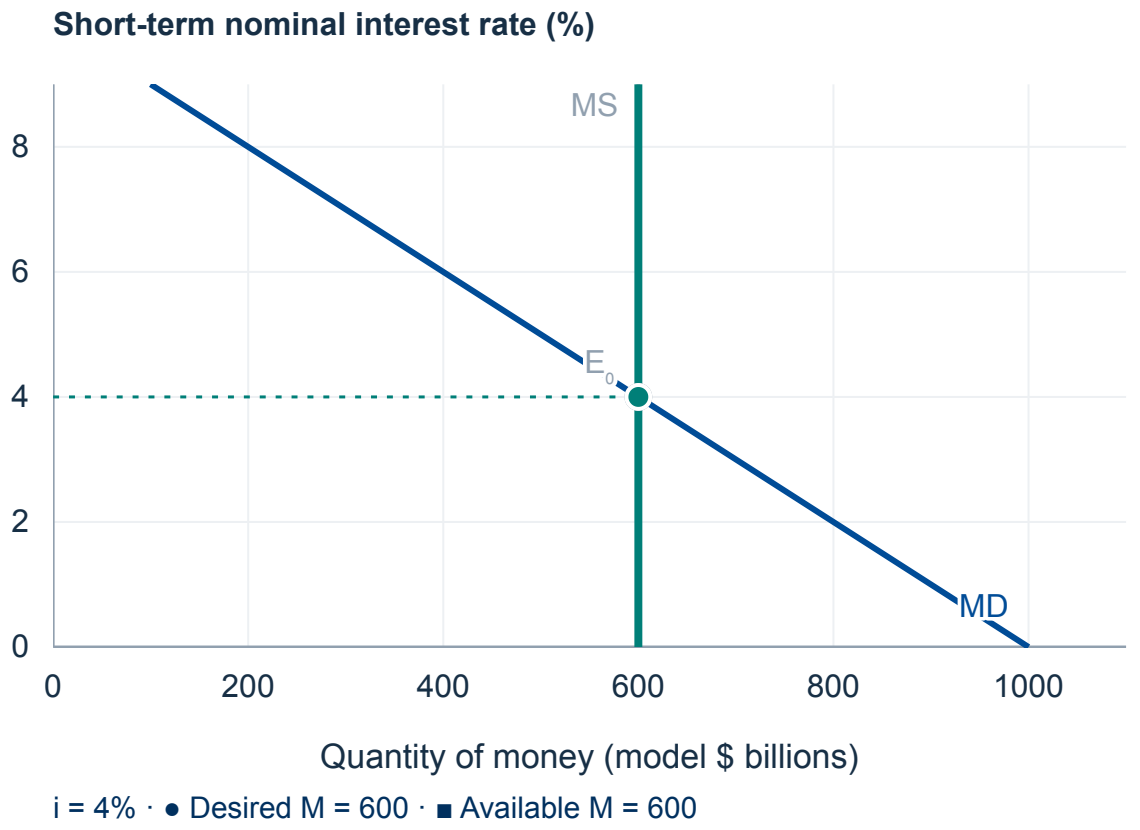
The bank can cover its $2,500 payment. It no longer needs the $500 overnight loan.

In the textbook model, more reserves can support new bank loans. New loans create deposits and raise the money supply.

The bond purchase adds reserves. New bank lending creates deposits.

Federal Reserve Board: implementation and IORB

Expansionary Monetary Policy



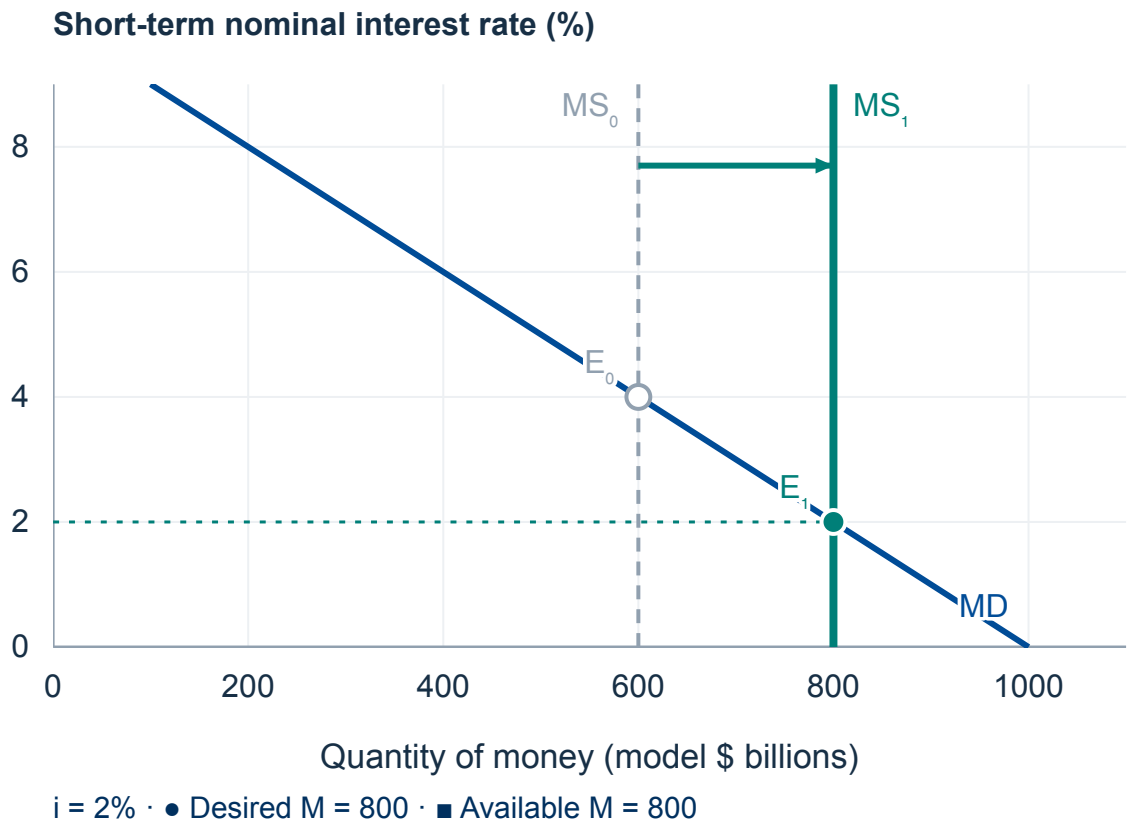
i = 4% · ● Desired M = 600 · ■ Available M = 600

Hold prices, real GDP and the return on money fixed.

Start

Start at M = 600 and i = 4%.

Expansionary Monetary Policy



Hold prices, real GDP and the return on money fixed.

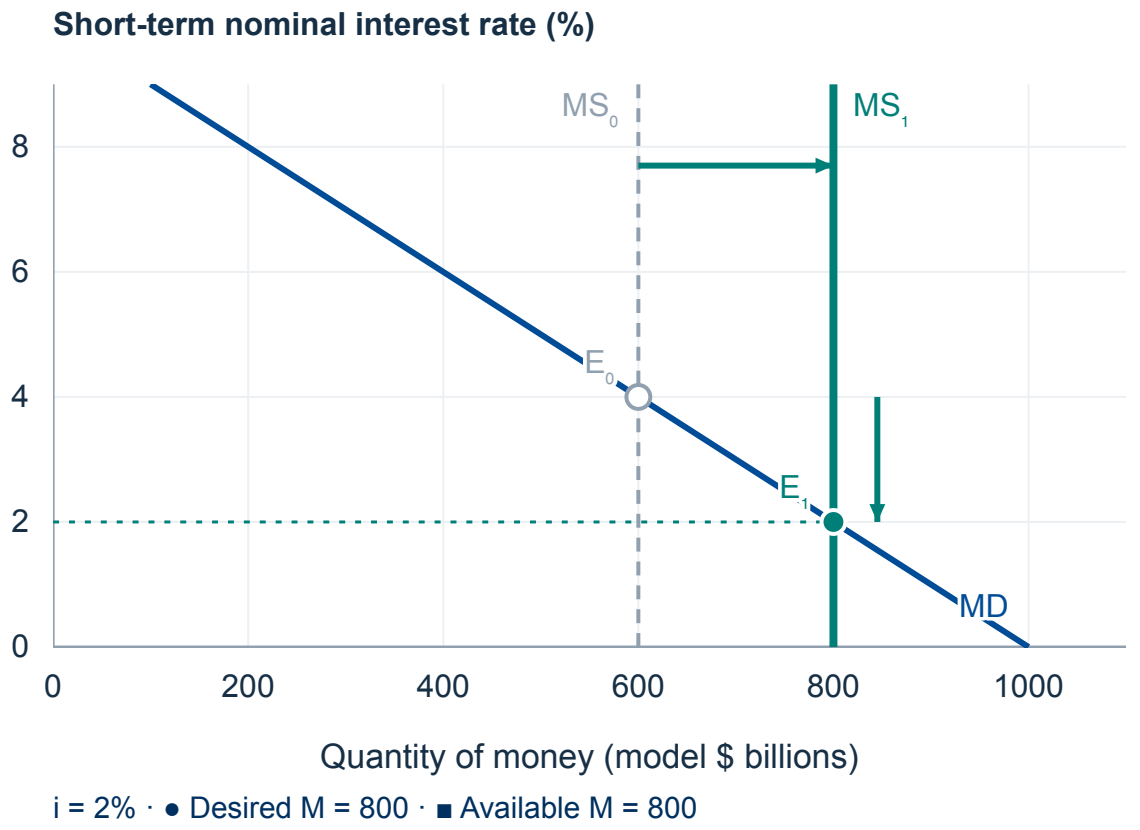
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More money

In this model, the Fed's purchase raises money supply to 800.

Expansionary Monetary Policy



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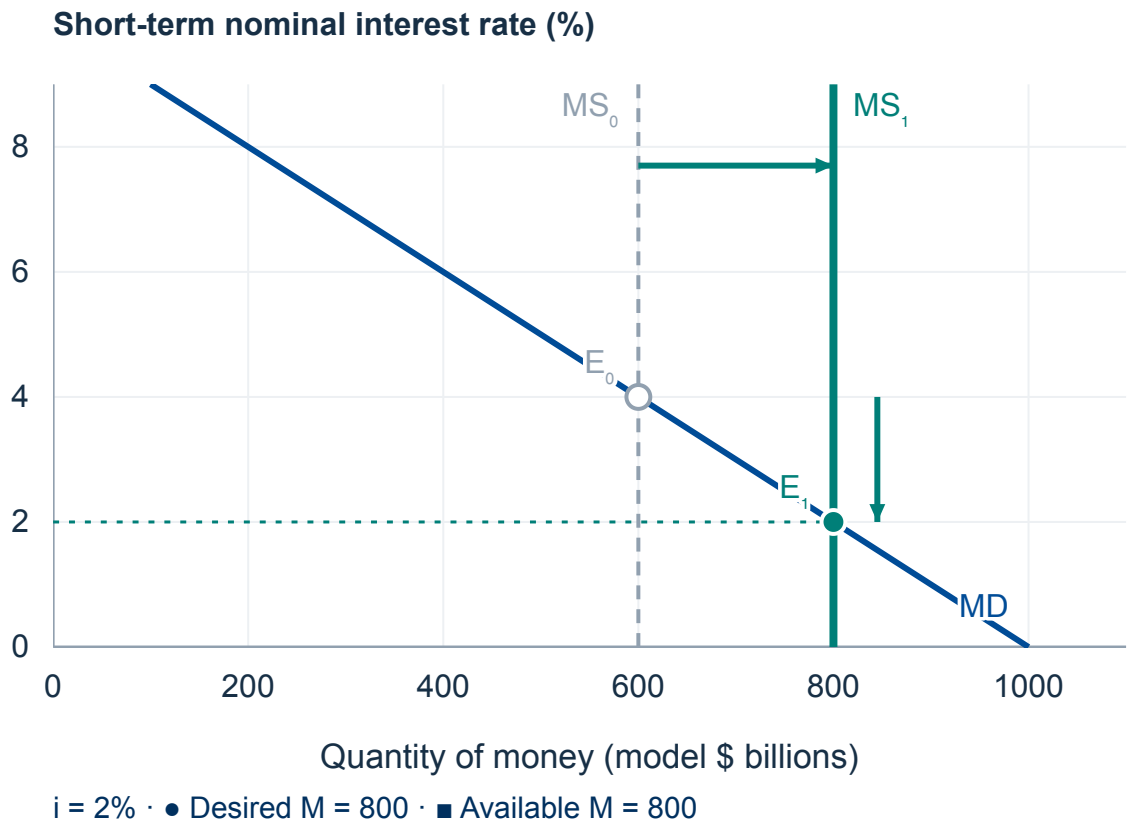
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Bond purchases

At 4%, people hold excess money and use it to buy bonds.

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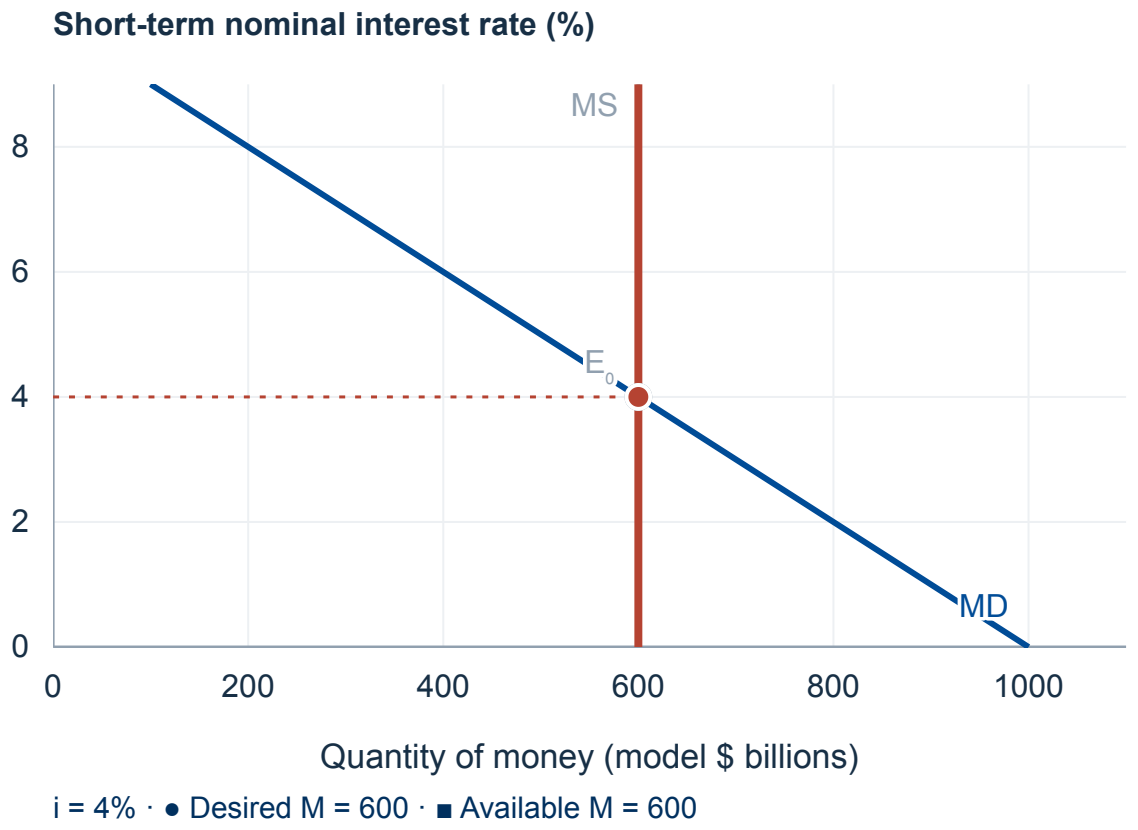
Bond purchases

At 4%, people hold excess money and use it to buy bonds.

Lower interest rate

Bond prices rise. The interest rate falls to 2%.

Contractionary Monetary Policy



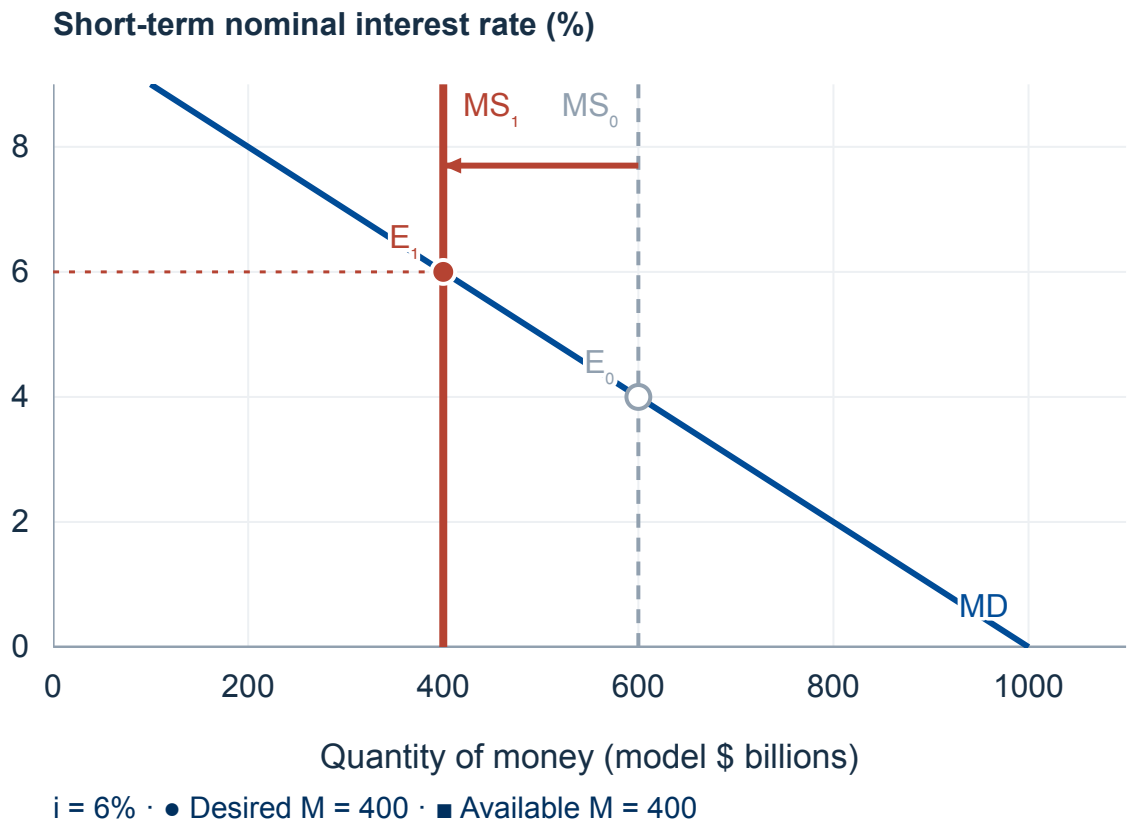
i = 4% · ● Desired M = 600 · ■ Available M = 600

Hold prices, real GDP and the return on money fixed.

Start

Start at M = 600 and i = 4%.

Contractionary Monetary Policy



i = 6% · ● Desired M = 400 · ■ Available M = 400

Hold prices, real GDP and the return on money fixed.

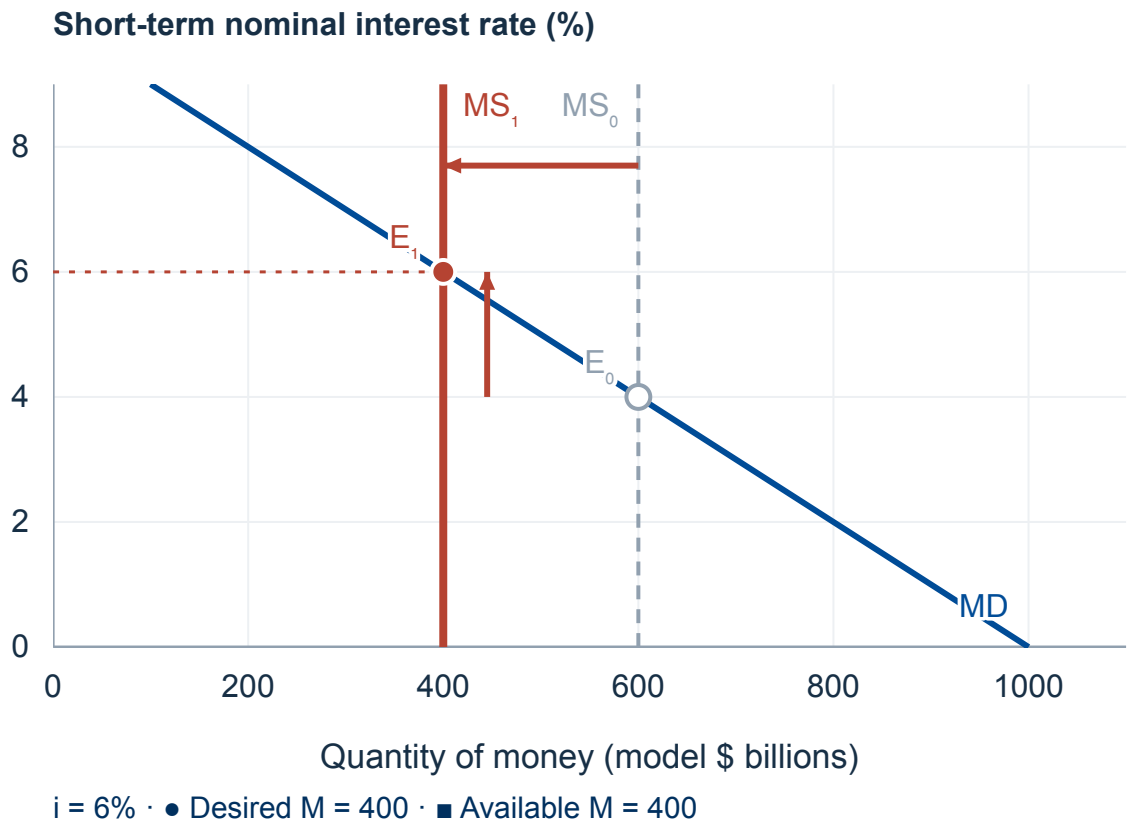
Start

Start at M = 600 and i = 4%.

Less money

In this model, a Fed sale lowers money supply to 400.

Contractionary Monetary Policy



Hold prices, real GDP and the return on money fixed.

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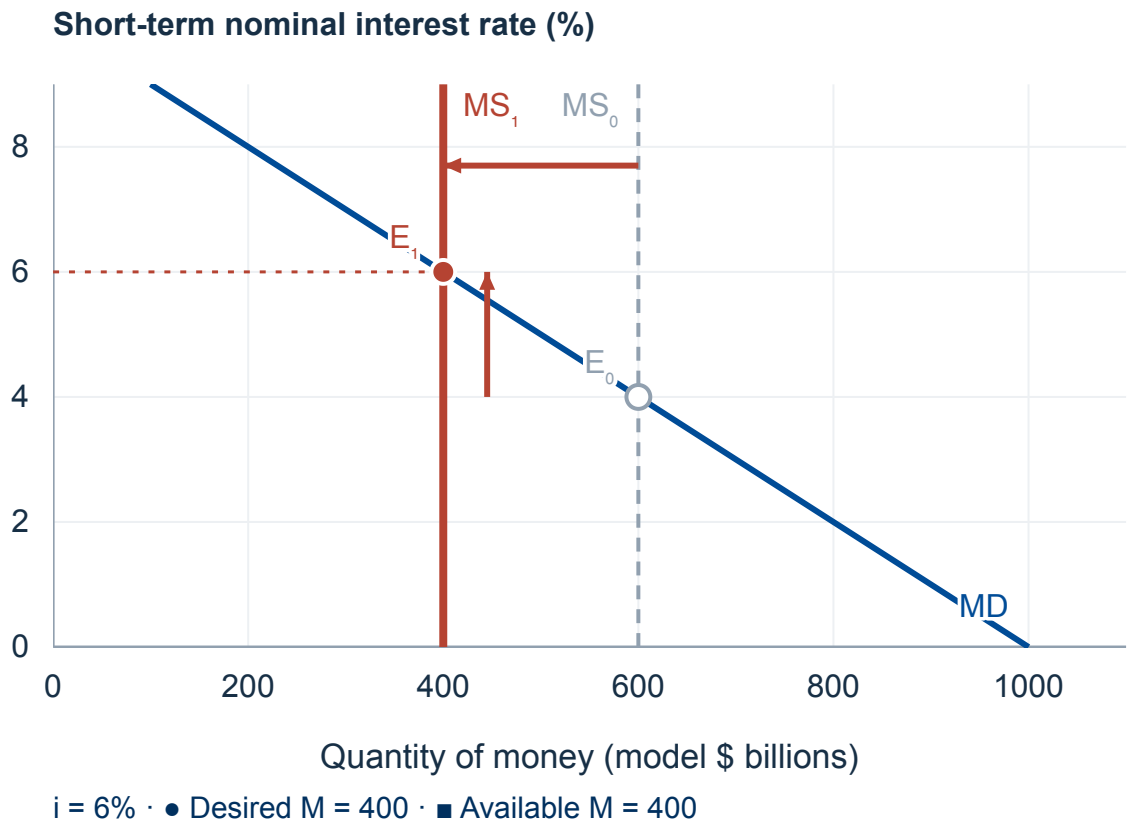
Less money

In this model, a Fed sale lowers money supply to 400.

Bond sales

At 4%, people want more money. They sell bonds to obtain it.

Contractionary Monetary Policy



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Start at M = 600 and i = 4%.

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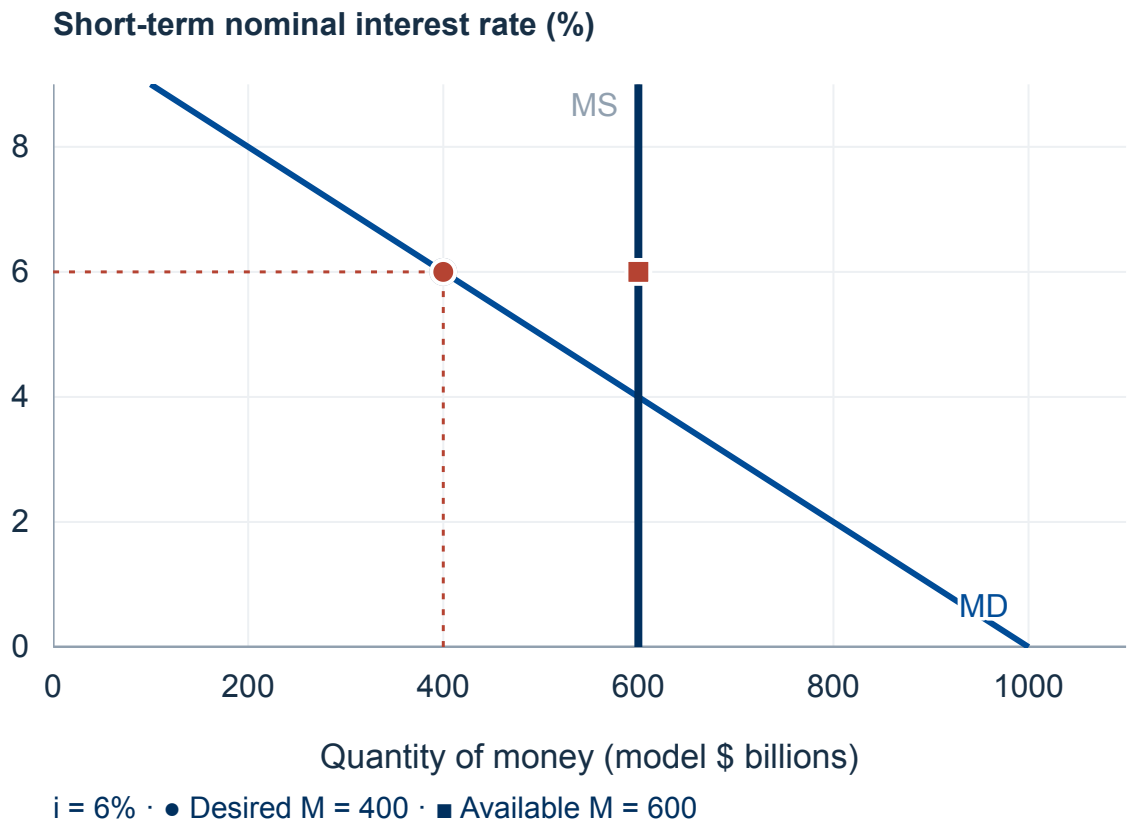
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Higher interest rate

Bond prices fall. The interest rate rises to 6%.

An Interest Rate above Equilibrium



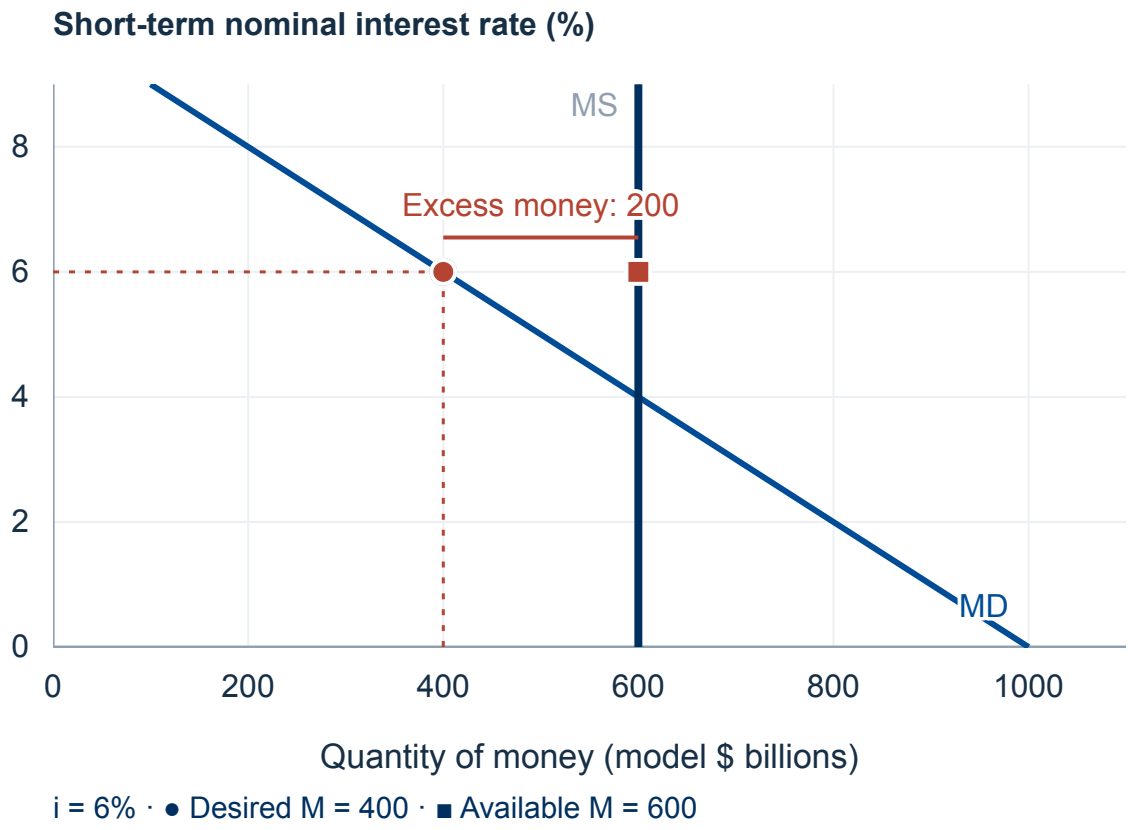
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Hold prices, real GDP and the return on money fixed.

At 6%

At 6%, people want 400 in money, but 600 is available.

An Interest Rate above Equilibrium



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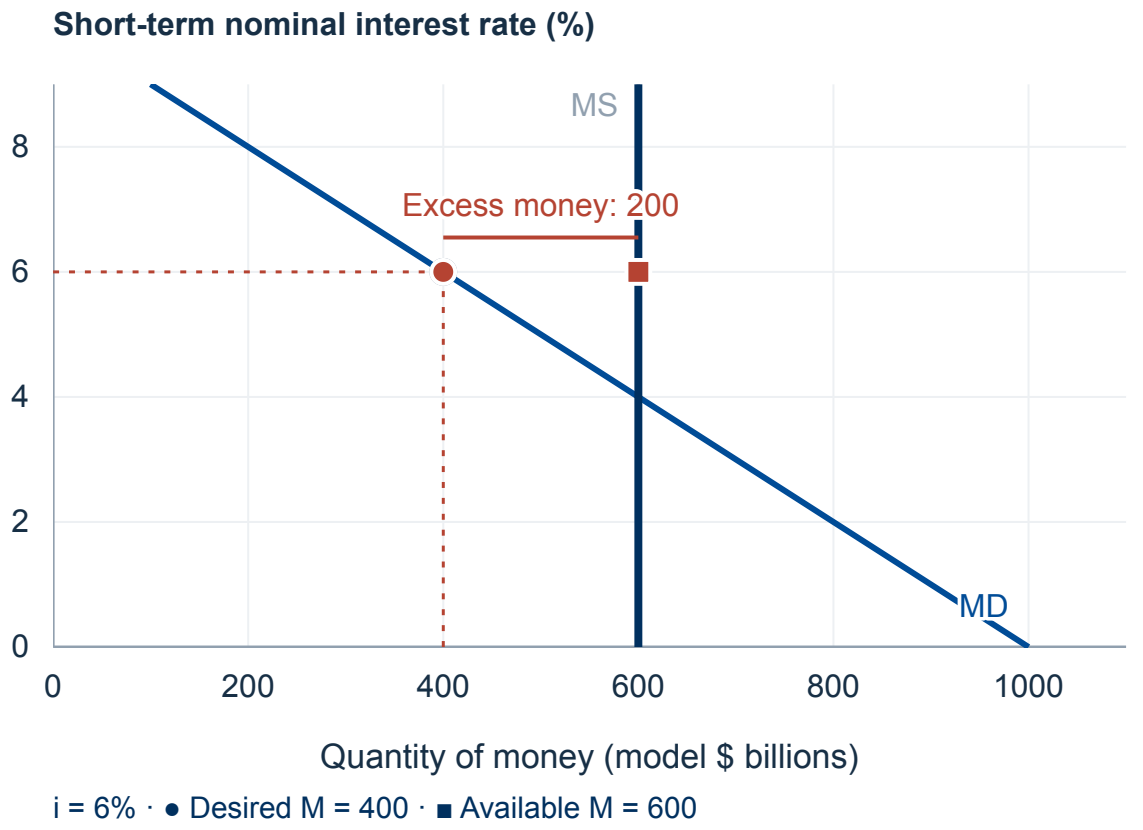
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Excess balances

They use excess money balances to buy bonds.

An Interest Rate above Equilibrium



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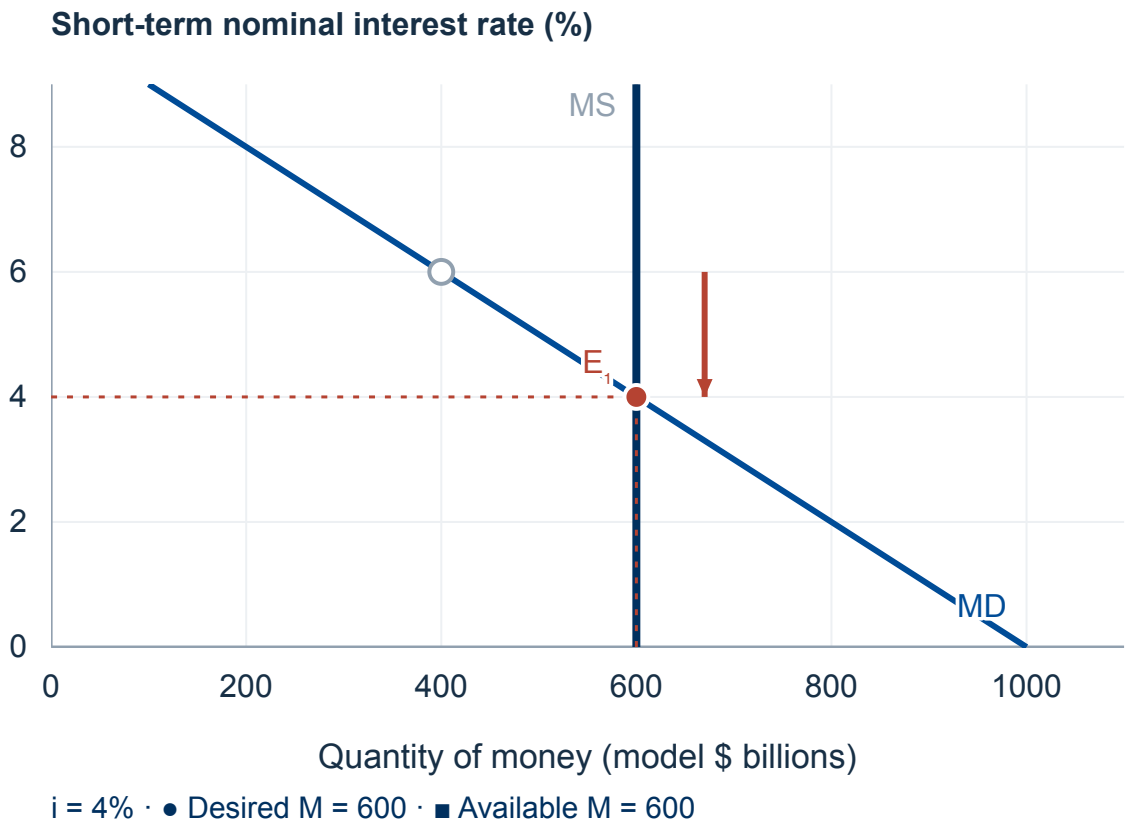
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Bond prices and yields

A bond paying $100 next year rises from $94.34 to $96.15. Its yield falls.

An Interest Rate above Equilibrium



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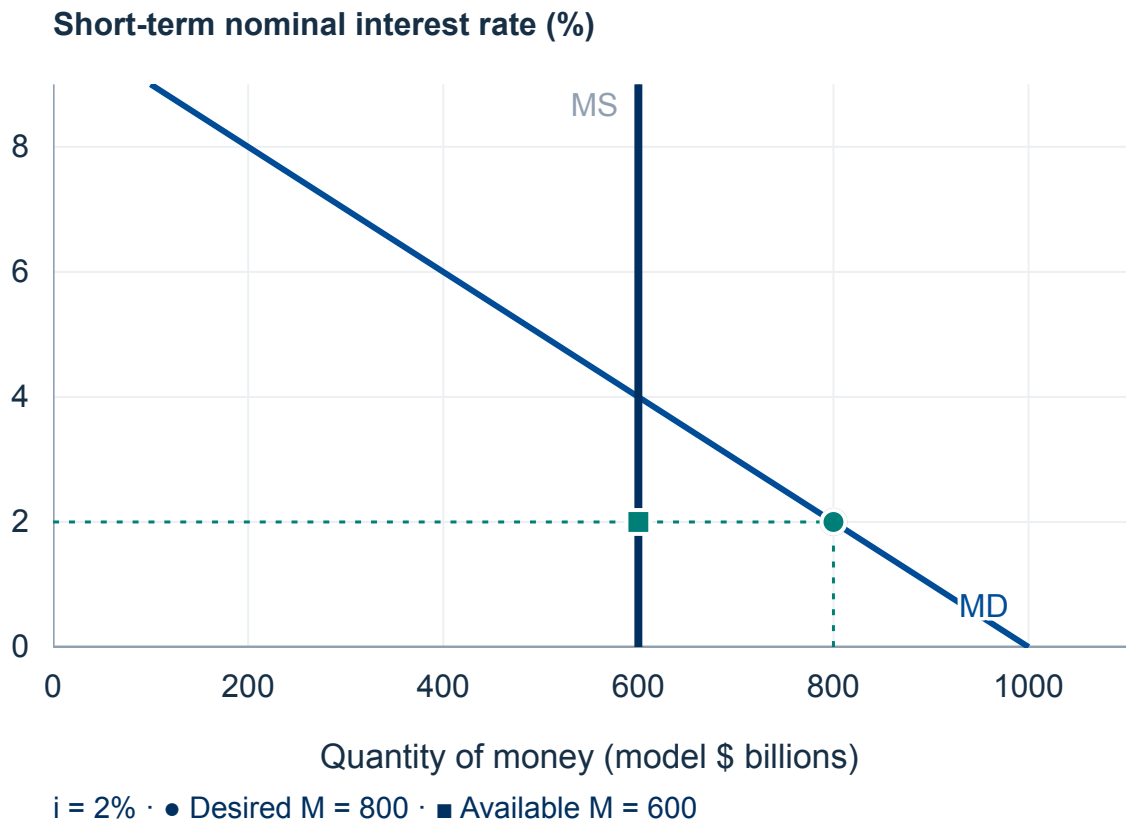
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At equilibrium

At 4%, money demand and money supply both equal 600.

An Interest Rate below Equilibrium

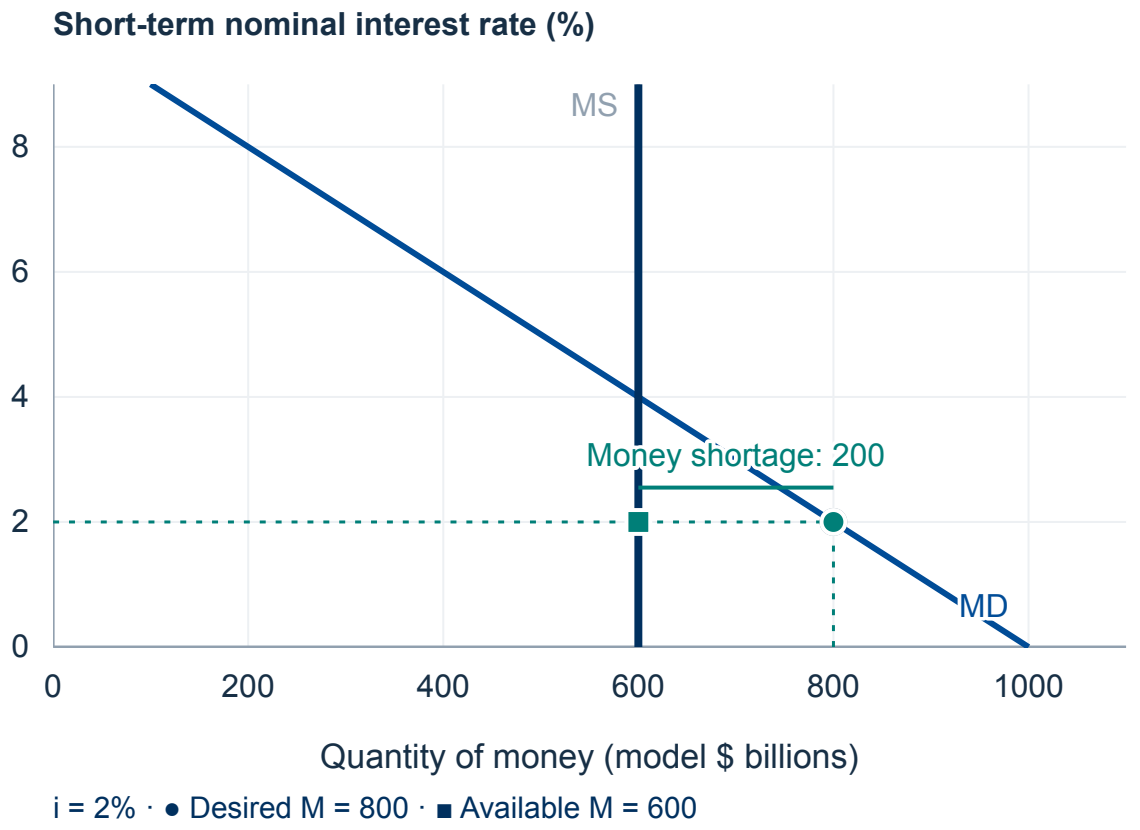


Hold prices, real GDP and the return on money fixed.

At 2%

At 2%, people want 800 in money, but only 600 is available.

An Interest Rate below Equilibrium



Hold prices, real GDP and the return on money fixed.

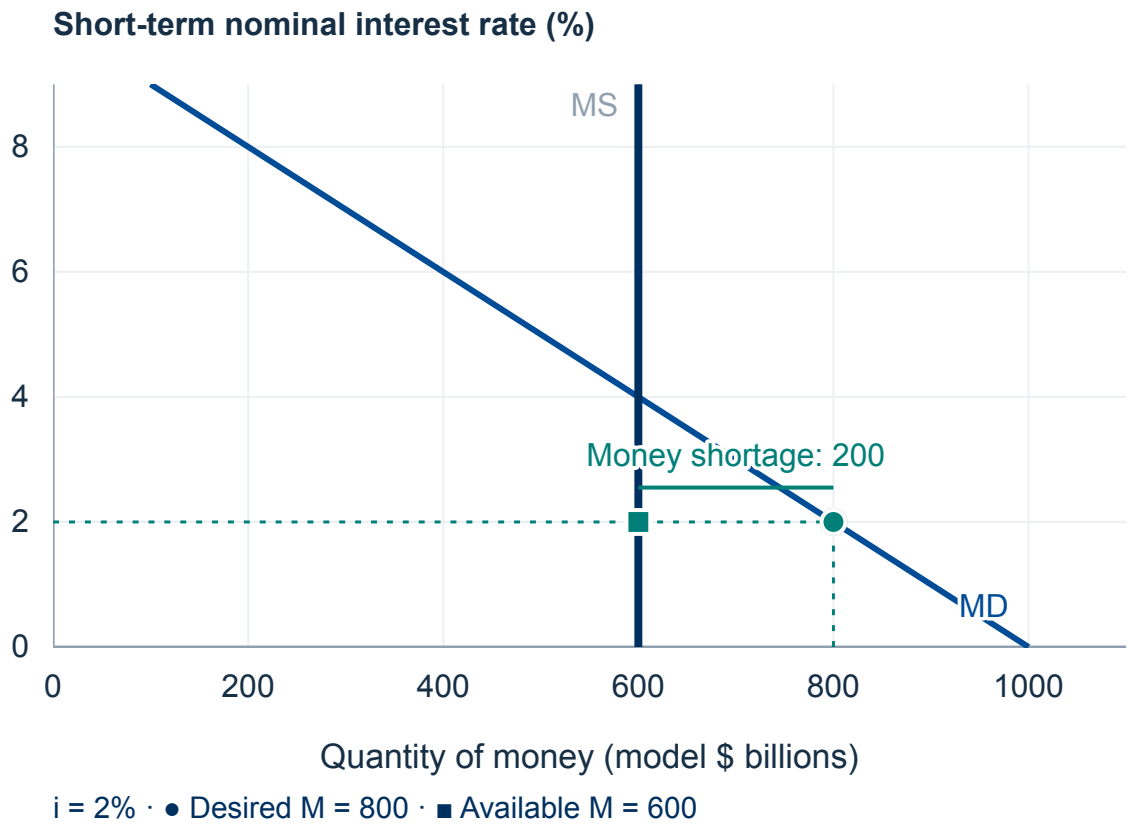
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A money shortage

They sell bonds to obtain money for payments.

An Interest Rate below Equilibrium



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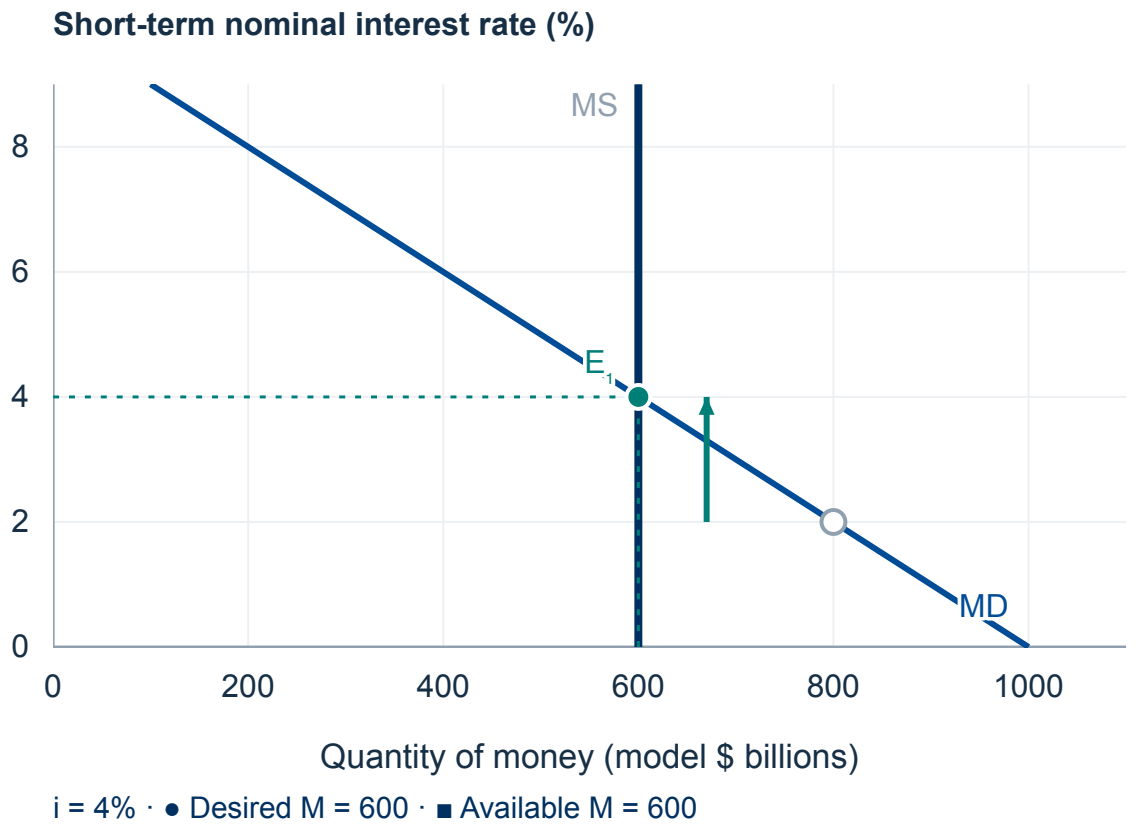
A money shortage

They sell bonds to obtain money for payments.

Bond prices and yields

A bond paying $100 next year falls from $98.04 to $96.15. Its yield rises.

An Interest Rate below Equilibrium



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Bond prices and yields

A bond paying $100 next year falls from $98.04 to $96.15. Its yield rises.

At equilibrium

At 4%, money demand and money supply both equal 600.

Money Supply and Interest Rates

Policy in the model	Money supply	Bond prices	Short-term rate
Fed buys bonds	Rises	Rise	Falls
Fed sells bonds	Falls	Fall	Rises

Hold prices, real GDP and the return on money fixed.

Money-Market Checkpoint

1. Money supply rises while MD stays fixed. What happens to the interest rate, and why?

2. The rate is below equilibrium. How does the money shortage disappear?

Money-Market Checkpoint

1. Money supply rises while MD stays fixed. What happens to the interest rate, and why?

> MS shifts right. People buy bonds with excess money, raising bond prices and lowering the rate.

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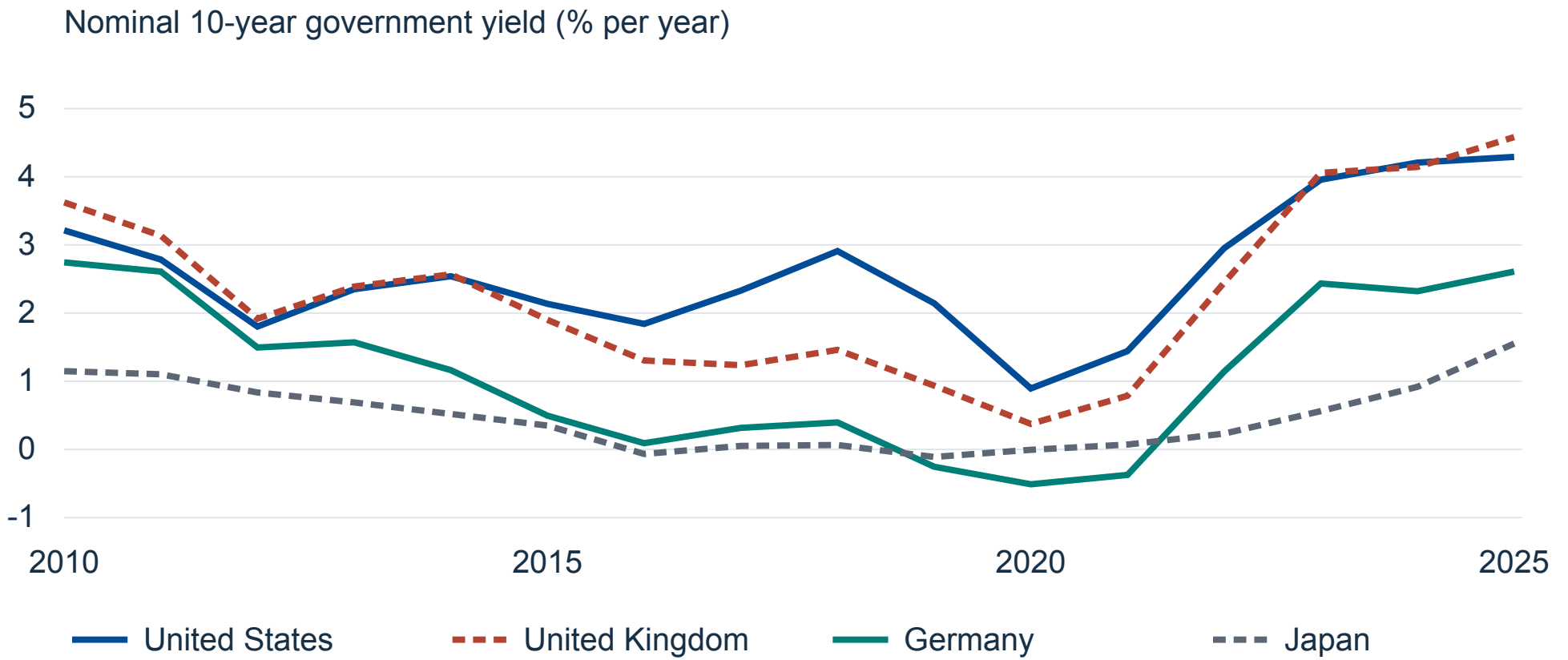
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> MS shifts right. People buy bonds with excess money, raising bond prices and lowering the rate.

2. The rate is below equilibrium. How does the money shortage disappear?

> People sell bonds. Bond prices fall and the rate rises, reducing the quantity of money demanded.

Long-Term Interest Rates across Countries



Annual averages, 2010–2025; own-currency yields. National benchmark definitions differ.

OECD (2026), Financial market statistics: long-term interest rates (accessed November 2, 2026) · OECD data terms, section 3

One-Year and Two-Year Investments

Invest $600 for two years	Year 1 rate	Year 2 rate
Two-year contract, annual compounding	5% fixed	5% fixed
Roll one-year investments: P5	4%	2% scenario
Roll one-year investments: P6	4%	8% scenario

- Answer P5 and P6 before revealing the payoffs.

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- P5: two years at 5% gives $661.50; rolling at 4% then 2% gives $636.48.
- P6: rolling at 4% then 8% gives $673.92, more than the fixed two-year contract.

Longer-Term Rates and Bond Prices

SVB securities sale · March 8, 2023	Amount
Securities sold	About $21 billion
Reported after-tax loss	About $1.8 billion

- Long-term rates reflect expected future short-term rates and compensation for risk.

Fixed bond payments do not rise when market interest rates rise.

Federal Reserve Board, SVB review (April 2023) · SEC Investor.gov: bonds

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- Long-term rates reflect expected future short-term rates and compensation for risk.
- When new bonds offer higher rates, older fixed-payment bonds fall in value.

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Securities sold	About $21 billion
Reported after-tax loss	About $1.8 billion

- Long-term rates reflect expected future short-term rates and compensation for risk.
- When new bonds offer higher rates, older fixed-payment bonds fall in value.
- SVB sold before maturity and reported a $1.8 billion after-tax loss.

Fixed bond payments do not rise when market interest rates rise.

Federal Reserve Board, SVB review (April 2023) · SEC Investor.gov: bonds

15.3 Monetary Policy and Aggregate Demand

- Interpret aggregate-demand shifts and demonstrate the effects of monetary policy.

Fiscal and Monetary Policy

Policy	Main tools	Useful for
Monetary policy	Interest rates and financial conditions	Changes in spending across the economy
Fiscal policy	Purchases, taxes and transfers	Direct spending or support for particular groups
Both together	A coordinated response	Severe downturns, especially when rates cannot fall much further

- New fiscal measures may need a law. Automatic stabilizers do not.

IMF: fiscal policy · Federal Reserve Board: policy goals and transmission

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- New fiscal measures may need a law. Automatic stabilizers do not.
- Adding demand does not repair a loss of productive capacity. Supply shocks create a policy tradeoff.

IMF: fiscal policy · Federal Reserve Board: policy goals and transmission

Monetary Policy and Aggregate Demand

Total planned spending = C + I + G + NX

1 The Fed lowers its policy rate. Borrowing rates for households and firms can fall.

Higher borrowing rates reverse this chain: C and I fall, and AD shifts left.

Federal Reserve Board: policy goals and transmission

Monetary Policy and Aggregate Demand

Total planned spending = C + I + G + NX

1. The Fed lowers its policy rate. Borrowing rates for households and firms can fall.
2. Cheaper car and appliance loans encourage household purchases: consumption (C) rises.

Higher borrowing rates reverse this chain: C and I fall, and AD shifts left.

Federal Reserve Board: policy goals and transmission

Monetary Policy and Aggregate Demand

Total planned spending = C + I + G + NX

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2. Cheaper car and appliance loans encourage household purchases: consumption (C) rises.
3. Cheaper business loans make more equipment and building projects worthwhile: investment (I) rises.

Higher borrowing rates reverse this chain: C and I fall, and AD shifts left.

Federal Reserve Board: policy goals and transmission

Monetary Policy and Aggregate Demand

Total planned spending = C + I + G + NX

1. The Fed lowers its policy rate. Borrowing rates for households and firms can fall.
2. Cheaper car and appliance loans encourage household purchases: consumption (C) rises.
3. Cheaper business loans make more equipment and building projects worthwhile: investment (I) rises.
4. At the same price level, higher C and I raise planned spending. AD shifts right.

Higher borrowing rates reverse this chain: C and I fall, and AD shifts left.

Federal Reserve Board: policy goals and transmission

Borrowing Costs and Investment

1 A bakery considers a $10,000 oven. Expected net receipts next year are $10,500.

These are business-loan rates, not the federal funds rate.

Federal Reserve Board: policy goals and transmission

Borrowing Costs and Investment

1. A bakery considers a $10,000 oven. Expected net receipts next year are $10,500.
2. Borrow at 6%: repay $10,600. The project loses $100.

These are business-loan rates, not the federal funds rate.

Federal Reserve Board: policy goals and transmission

Borrowing Costs and Investment

1. A bakery considers a $10,000 oven. Expected net receipts next year are $10,500.
2. Borrow at 6%: repay $10,600. The project loses $100.
3. Borrow at 4%: repay $10,400. The project gains $100.

These are business-loan rates, not the federal funds rate.

Federal Reserve Board: policy goals and transmission

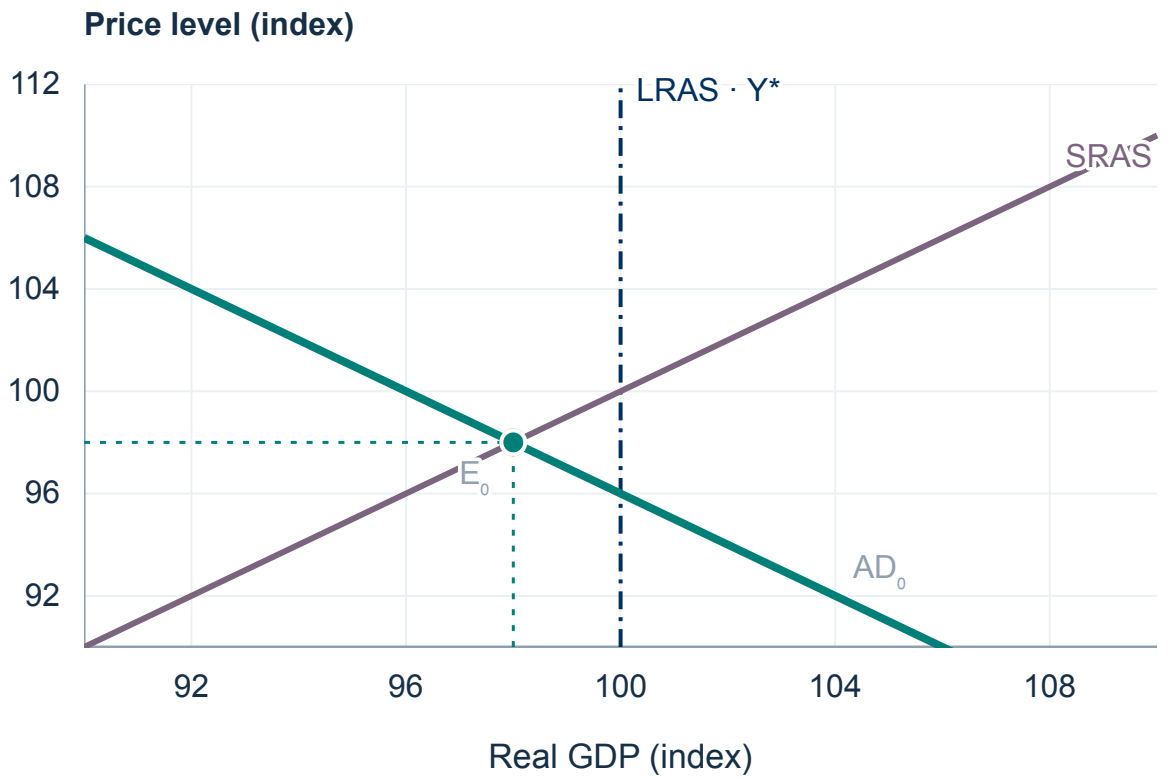
Borrowing Costs and Investment

1. A bakery considers a $10,000 oven. Expected net receipts next year are $10,500.
2. Borrow at 6%: repay $10,600. The project loses $100.
3. Borrow at 4%: repay $10,400. The project gains $100.
4. Cheaper borrowing can raise investment and consumption. AD shifts right.

These are business-loan rates, not the federal funds rate.

Federal Reserve Board: policy goals and transmission

Expansionary Policy and Aggregate Demand



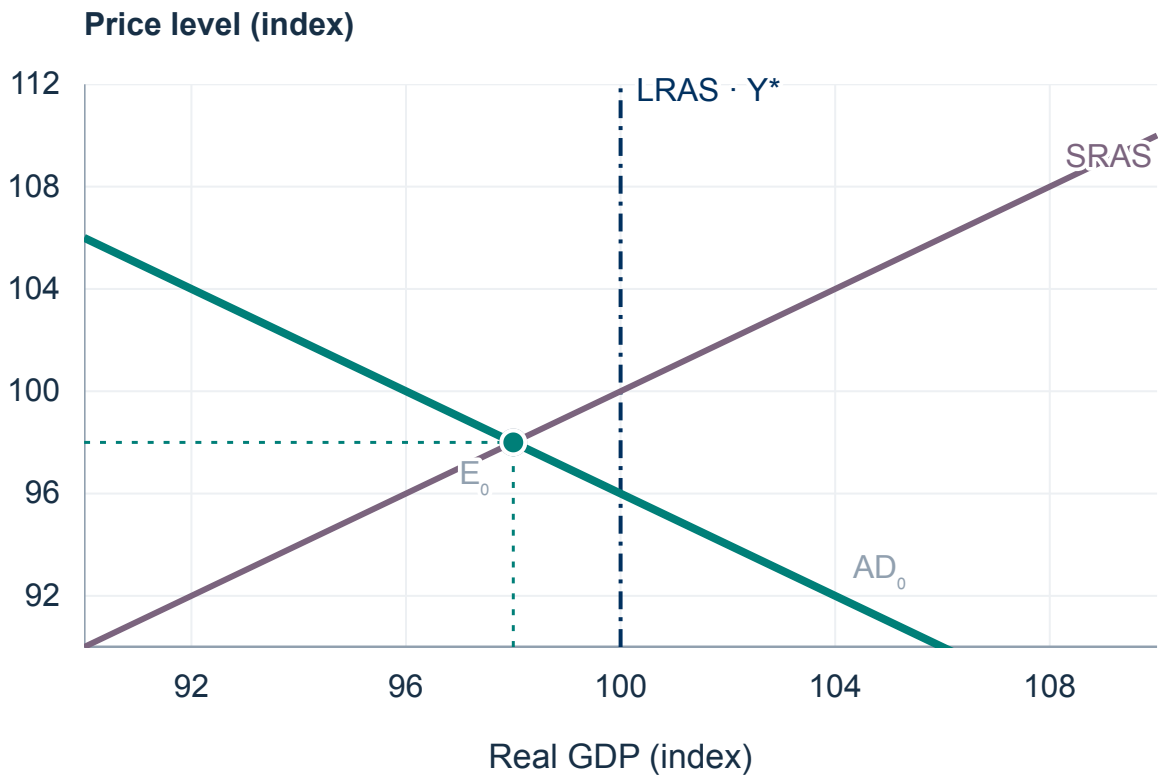
Y = 98 · P = 98 · SRAS and potential output fixed

SRAS and potential output stay fixed.

Below potential

Output is 98; potential output is 100.

Expansionary Policy and Aggregate Demand



Y = 98 · P = 98 · SRAS and potential output fixed

SRAS and potential output stay fixed.

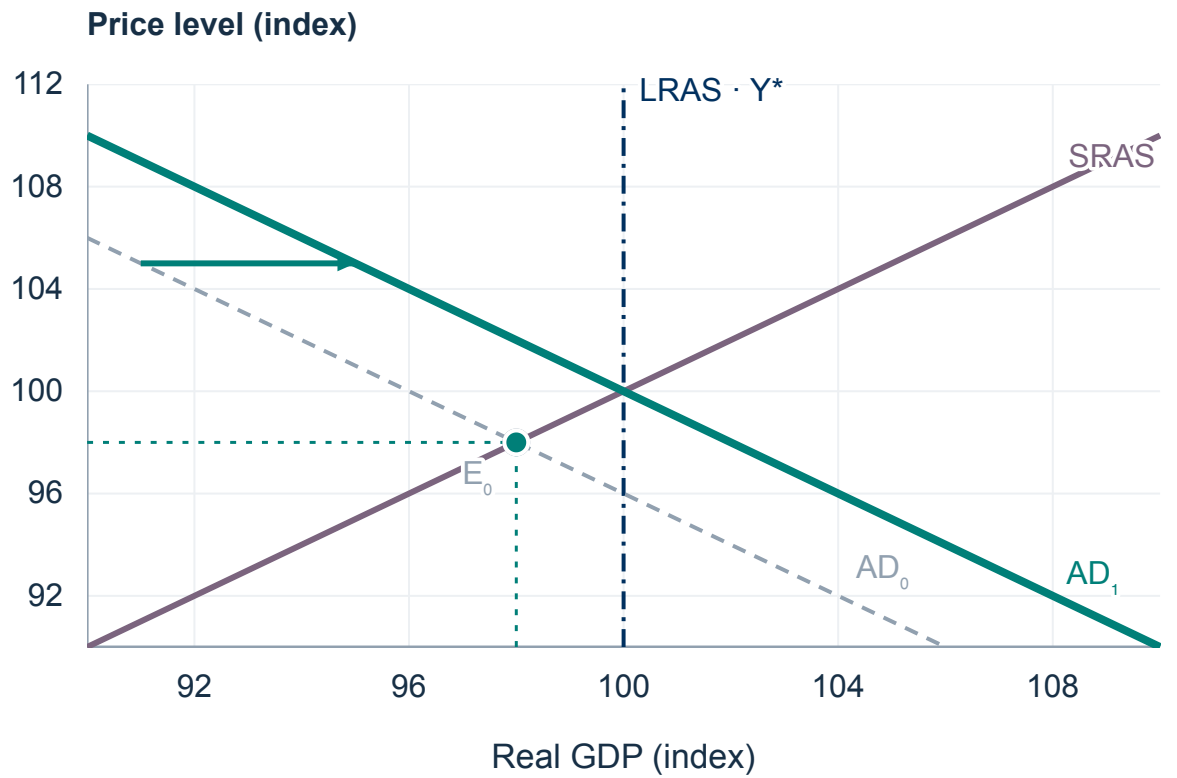
Below potential

Output is 98; potential output is 100.

Spending rises

Lower borrowing costs encourage C and I.

Expansionary Policy and Aggregate Demand



Y = 98 · P = 98 · SRAS and potential output fixed

SRAS and potential output stay fixed.

Below potential

Output is 98; potential output is 100.

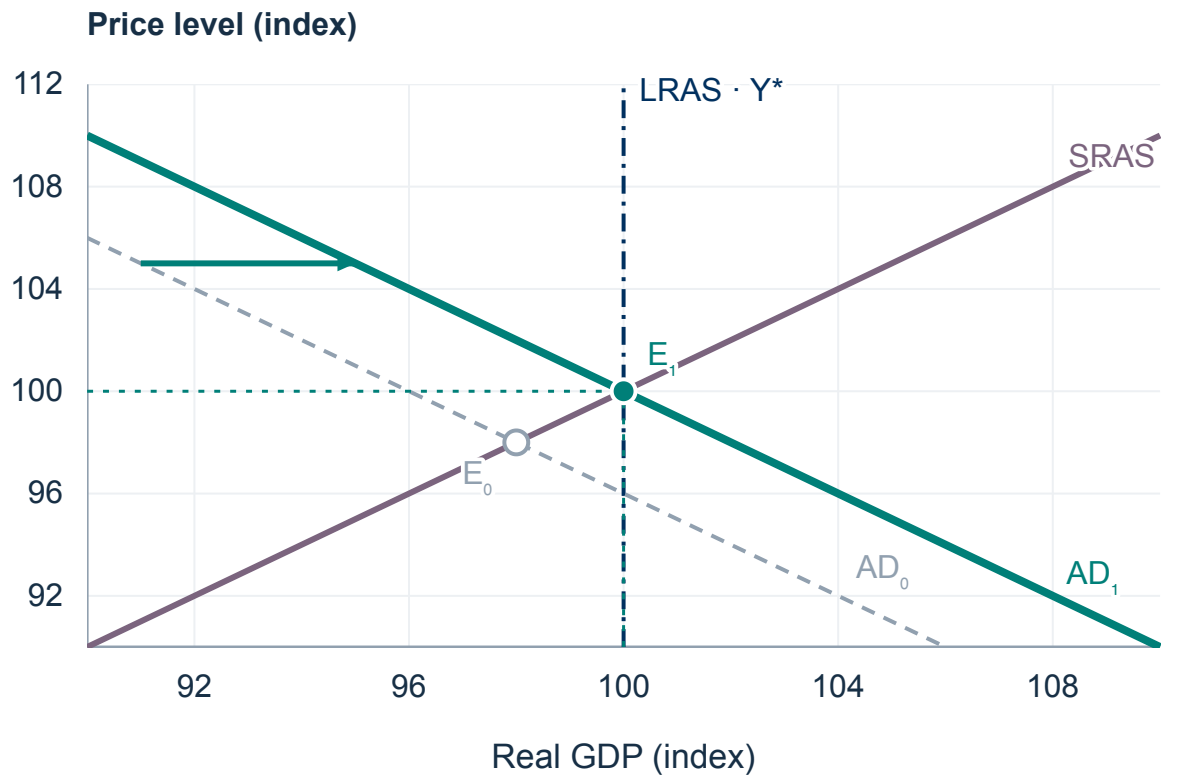
Spending rises

Lower borrowing costs encourage C and I.

AD shifts right

AD shifts right. SRAS and potential output stay fixed.

Expansionary Policy and Aggregate Demand



Y = 100 · P = 100 · SRAS and potential output fixed

SRAS and potential output stay fixed.

Below potential

Output is 98; potential output is 100.

Spending rises

Lower borrowing costs encourage C and I.

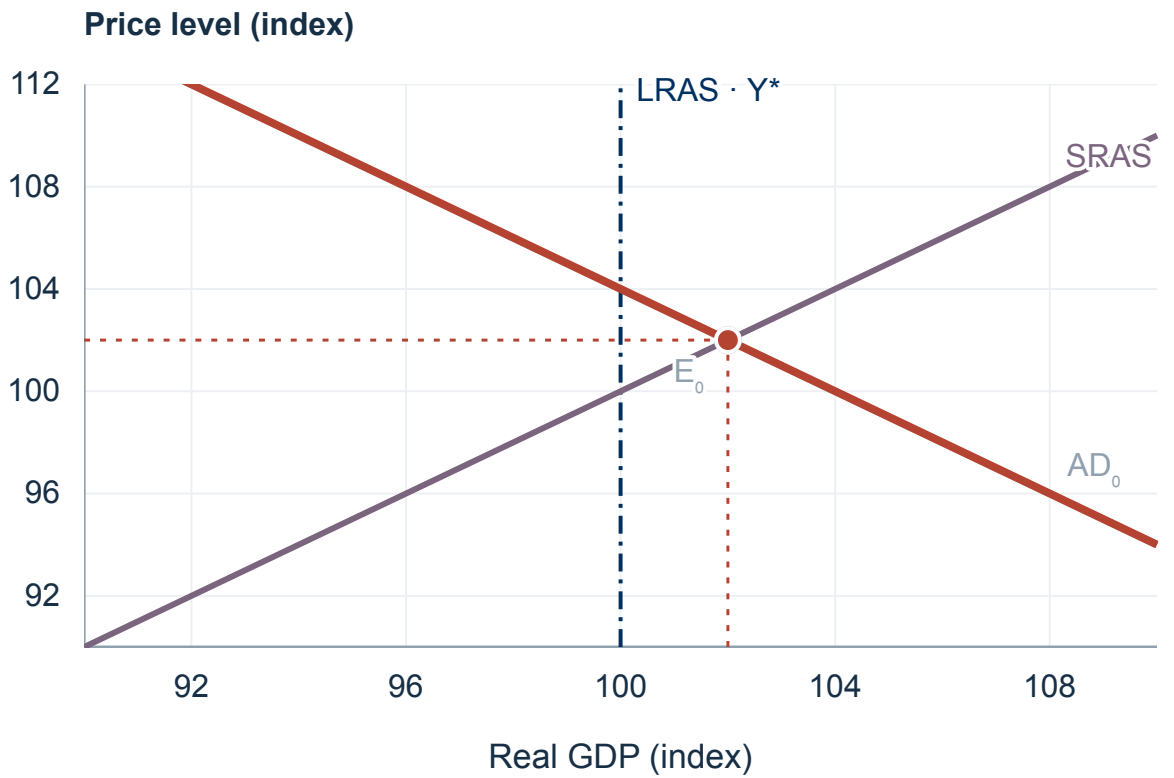
AD shifts right

AD shifts right. SRAS and potential output stay fixed.

New equilibrium

The new equilibrium has output 100 and price level 100.

Contractionary Policy and Aggregate Demand



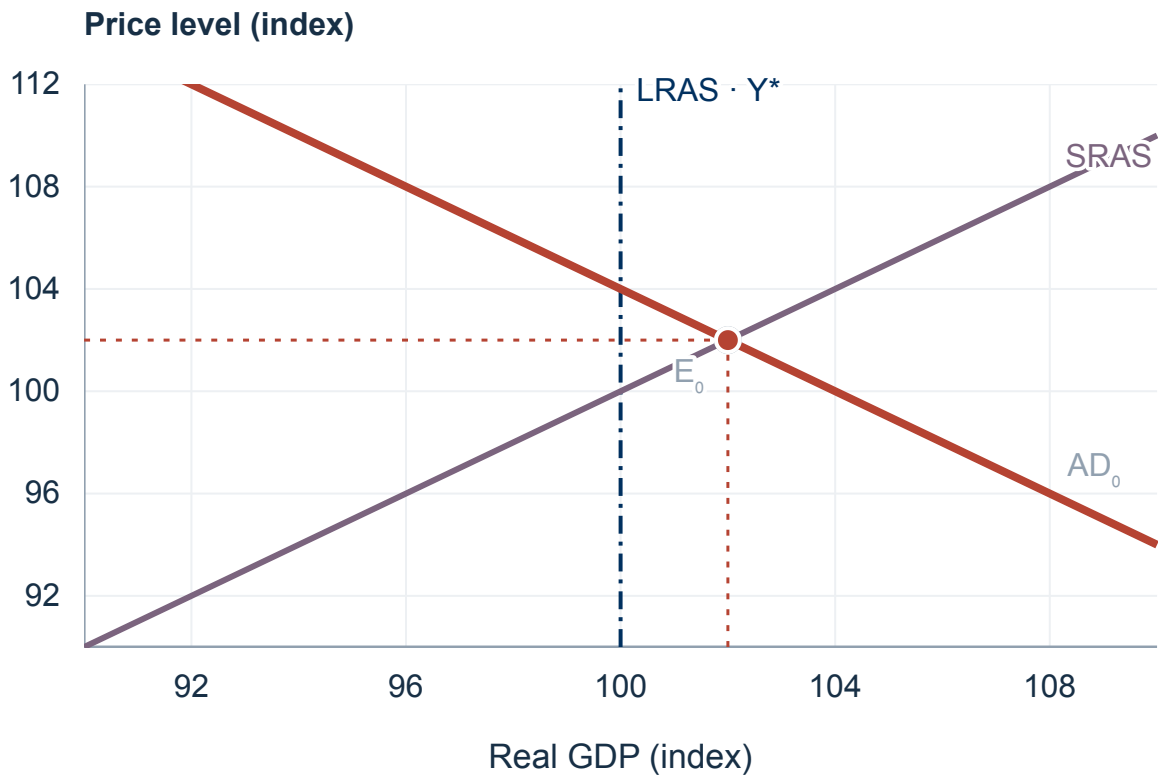
Y = 102 · P = 102 · SRAS and potential output fixed

SRAS and potential output stay fixed.

Above potential

Output is 102; potential output is 100.

Contractionary Policy and Aggregate Demand



Y = 102 · P = 102 · SRAS and potential output fixed

SRAS and potential output stay fixed.

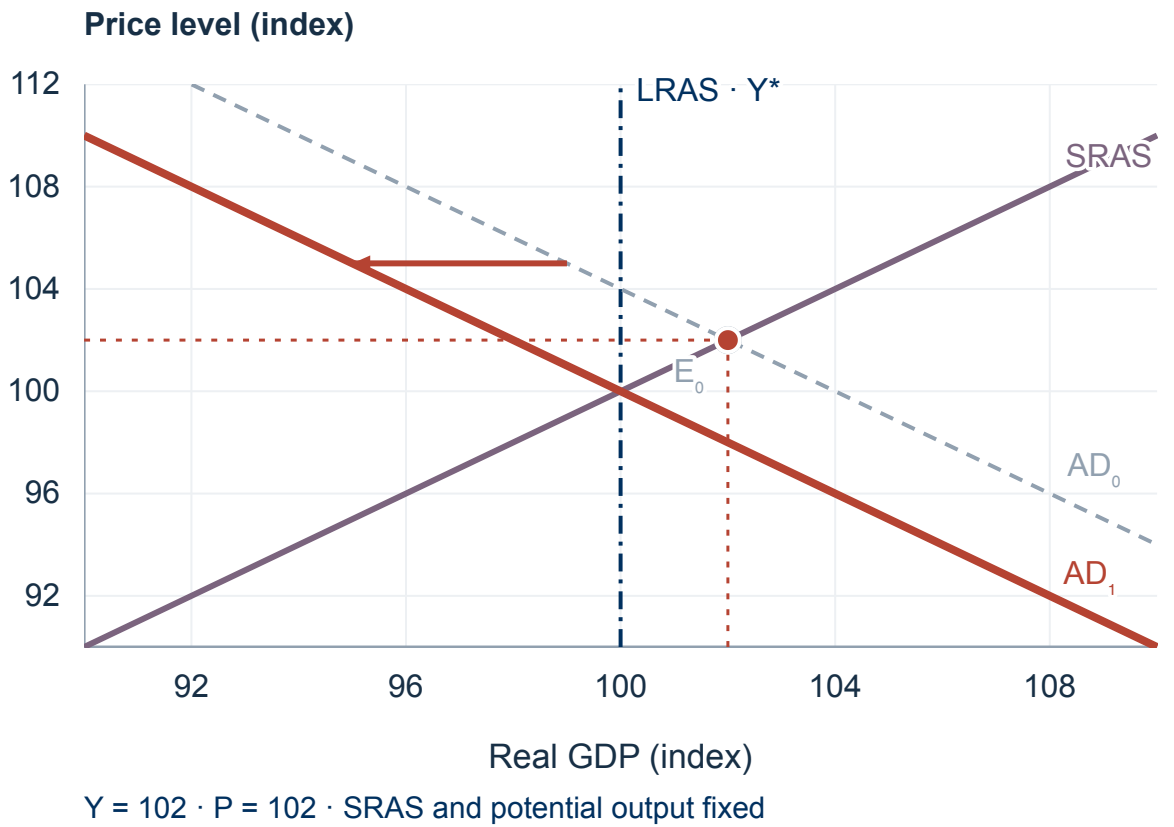
Above potential

Output is 102; potential output is 100.

Spending falls

Higher borrowing costs reduce C and I.

Contractionary Policy and Aggregate Demand



Y = 102 · P = 102 · SRAS and potential output fixed

SRAS and potential output stay fixed.

Above potential

Output is 102; potential output is 100.

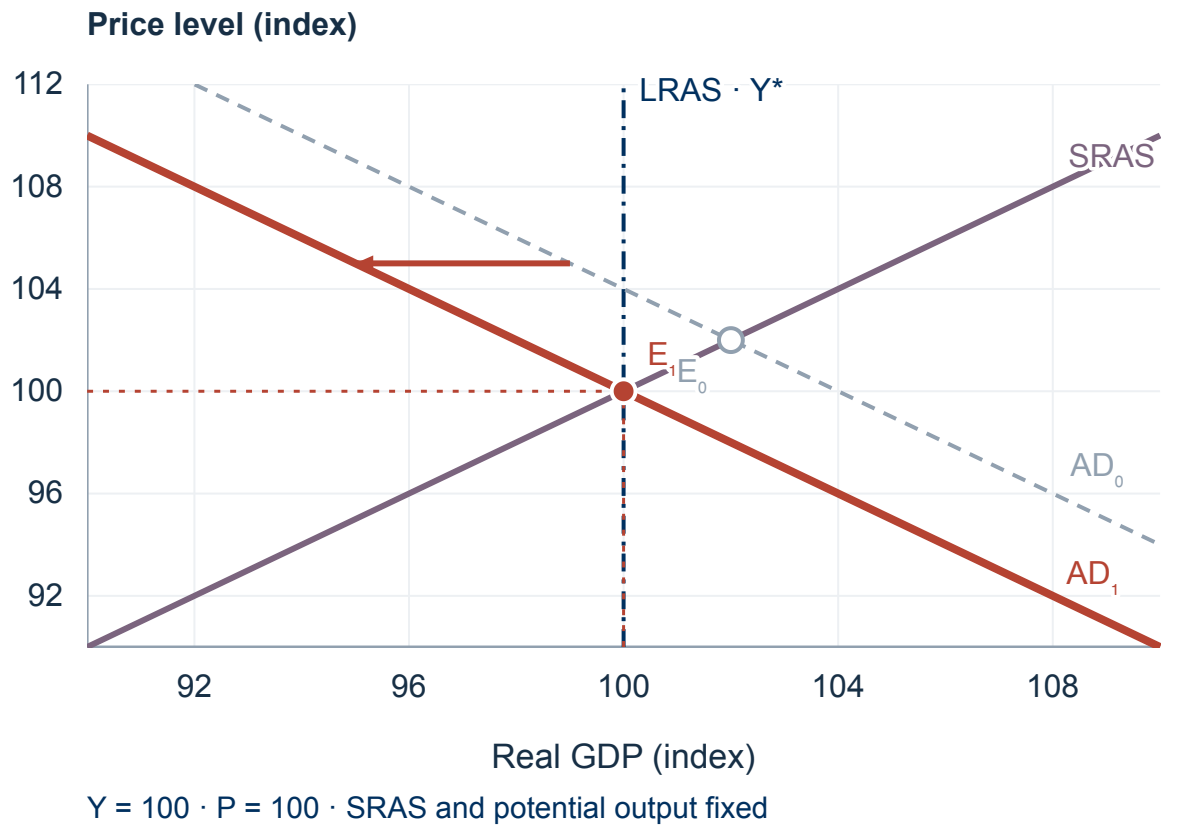
Spending falls

Higher borrowing costs reduce C and I.

AD shifts left

AD shifts left. SRAS and potential output stay fixed.

Contractionary Policy and Aggregate Demand



SRAS and potential output stay fixed.

Above potential

Output is 102; potential output is 100.

Spending falls

Higher borrowing costs reduce C and I.

AD shifts left

AD shifts left. SRAS and potential output stay fixed.

New equilibrium

Output returns to 100. The price level is lower than without the policy.

Monetary Policy: Short-Run Effects

Channel	Expansionary policy	Contractionary policy
Money-market model	MS right; interest rate falls	MS left; interest rate rises
Borrowing costs	Can fall	Can rise
Spending	C and I rise; AD right	C and I fall; AD left
With SRAS fixed	Output and price level rise	Output and price level are lower

Potential output stays fixed in these short-run comparisons.